

V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning

META

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Introduction

Motivation: Why World Models?

- Humans can:
 - Adapt to new tasks
 - Predict effects of actions
 - Perform goal-directed planning
- AI agents should similarly:
 - **Understand** physical world
 - **Predict** future states
 - **Plan** in unseen environments

Introduction

Previous approaches and limitations:

1. Classical predictive world models

- Require: state + action + reward (interaction data)
- Limitation: poor scalability

2. Video generation-based models

- Trained on internet-scale video
- Limitation:
 - Weak at robot control/planning
 - Expensive
 - Limited evaluation of planning (focusing on prediction/visual quality)

- Existing methods fail to scale to both representation learning & control

Introduction

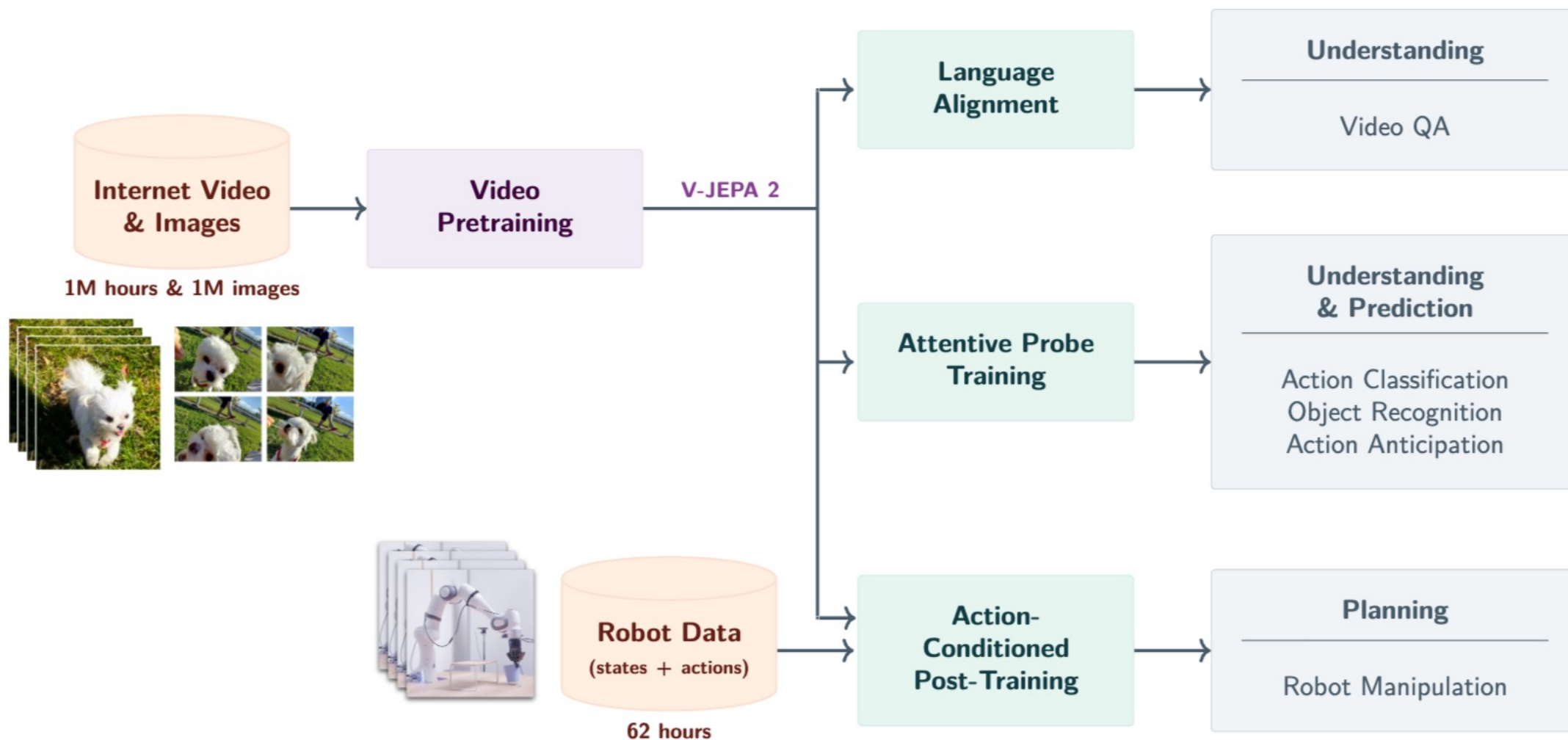
Key Idea: **Self-supervised World Model** from video

- Hypothesis: World knowledge can be learned from observation

Method Backbone: **JEPA (Joint-Embedding Predictive Architecture)**

- Predict in representation space (not pixels)
- Contrast:
 - Video generation → pixel-level prediction
 - JEPA → **semantic-level prediction**

V-JEPA 2 Overview



V-JEPA 2 Overview

1. Self-Supervised Pretraining (V-JEPA 2)

- Input: internet-scale video (action-free)
- Task: mask-denoising feature prediction

2. Action-Condition World Model (V-JEPA 2-AC)

- Small interaction dataset
- Learns:
 - Next state representation prediction
 - Conditioned on action + state

3. Planning

- Uses learned world model

V-JEPA2: Self-Supervised Video Pretraining

Key Idea:

- Learn video representation from internet-scale video (1M hours)
- No labels, no actions → self-supervised learning

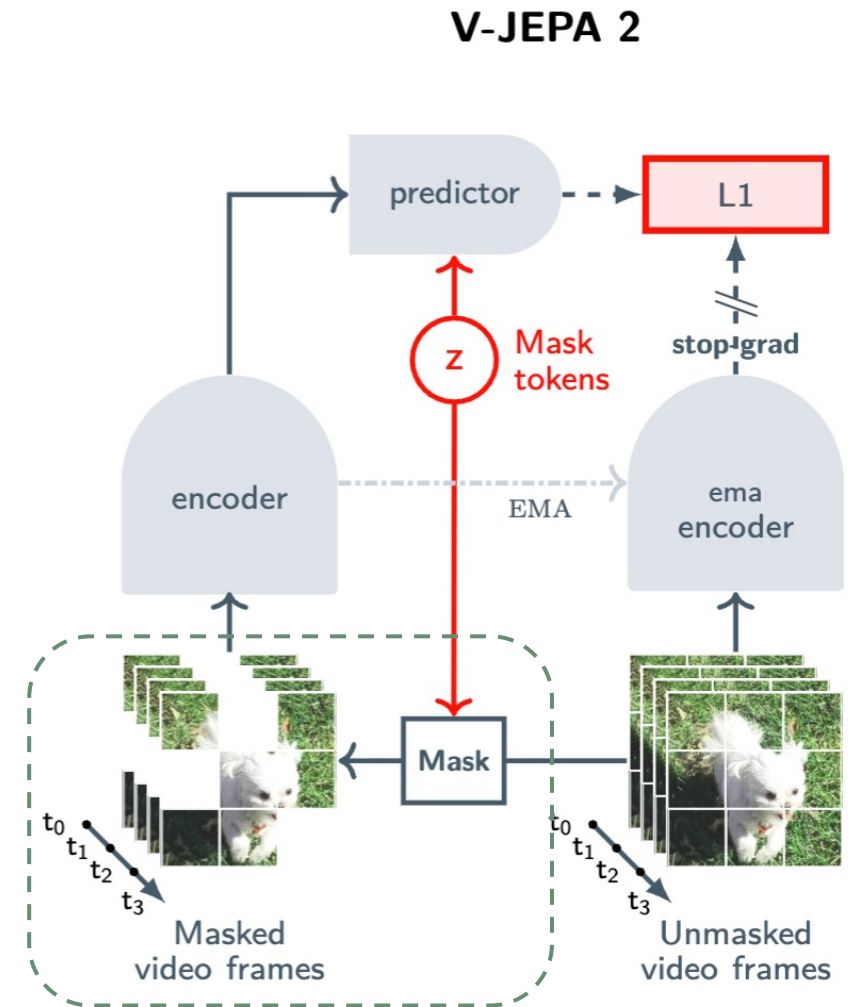
Training Objective:

- Masked video patches → **predict representation space**
- Not pixel reconstruction

V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

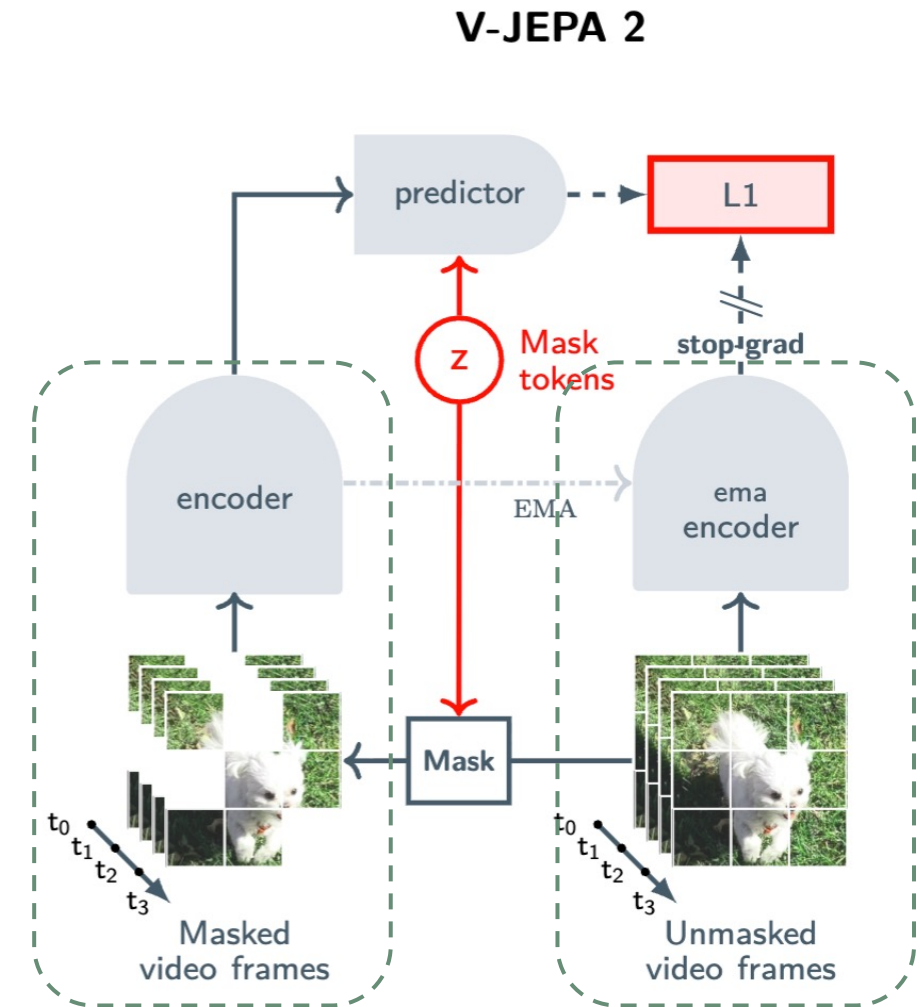
- (1) Patchify
 - Video clip $\rightarrow$ sequence of tokens (tubelets)
- (2) Masking
 - Randomly drop a subset of tokens
- (3) Encoder
 - Extract representations from visible patches
- (4) Predictor
 - Input: visible representations + mask tokens
 - Output: predicted representation of masked patches
- (5) EMA Encoder
 - Extract target representations from the original video



V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

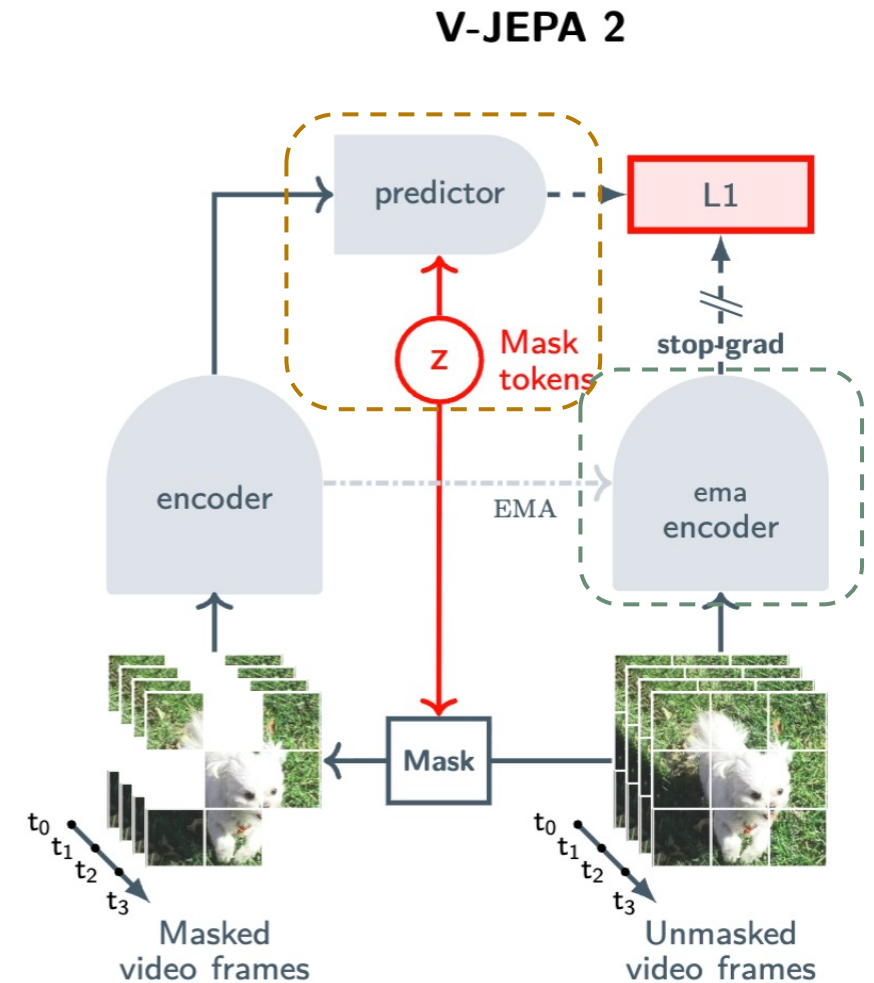
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V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

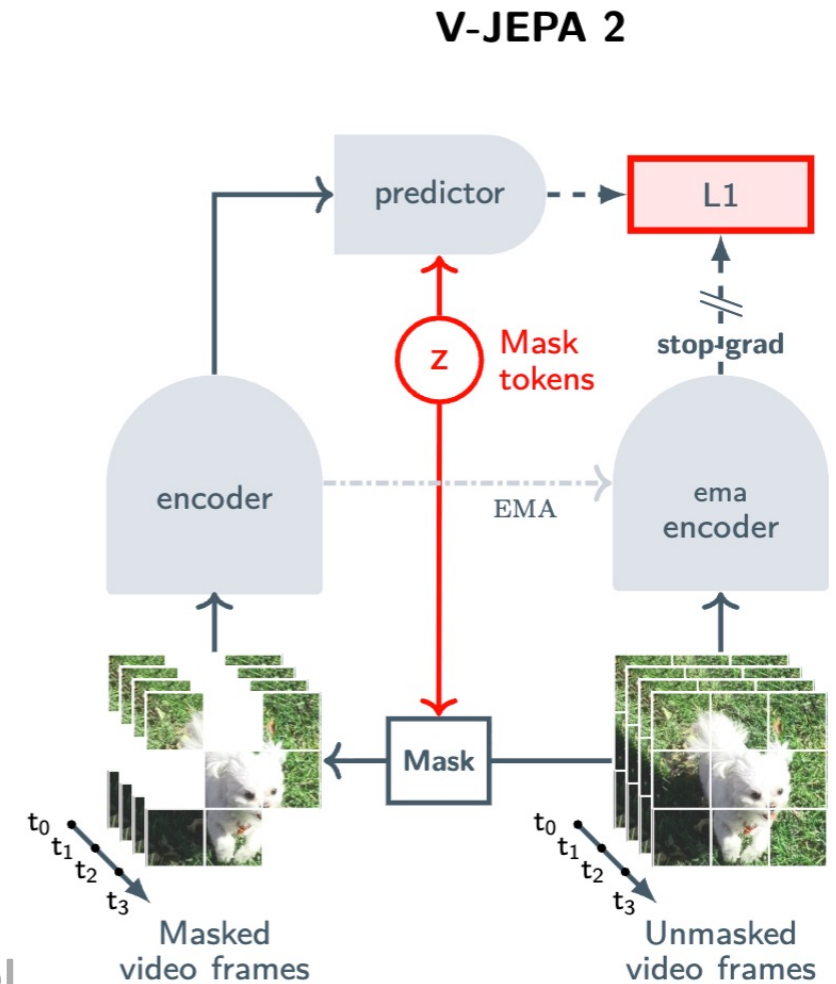
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V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

- (1) Patchify
 - Video clip $\rightarrow$ sequence of tokens (tubelets)
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 - Randomly drop a subset of tokens
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 - Extract representations from visible patches
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 - Input: visible representations + mask tokens
 - Output: **predicted representation** of masked patches
 - (5) EMA Encoder
 - Extract **target representations** from the original video
- Predict **missing representations** instead of reconstructing pixel



V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

- Goal: Predict the representation(y) of the original video from a masked view(x)
- Loss:

$$\underset{\theta, \phi, \Delta_y}{\text{minimize}} \left\| P_{\phi} \left(\Delta_y, E_{\theta}(x) \right) - \text{sg} \left(E_{\bar{\theta}}(y) \right) \right\|_1$$

Δ_y : Mask tokens (location of dropped patches)

E_{θ} : Encoder

$E_{\bar{\theta}}$: EMA Encoder

P_{ϕ} : Predictor

$\text{sg}(\cdot)$: stop gradient operation

- Loss is applied only to **masked patches**

V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

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- Preventing Representation Collapse
 - Stop-gradient blocks gradient flow through the target branch
 - EMA encoder provides stable target representations
 - EMA: $\bar{\theta} = m\bar{\theta} + (1 - m)\theta$

V-JEPA2: Self-Supervised Video Pretraining

Scaling Ingredients

- Data: 2M $\rightarrow$ 22M videos
- Model: 300M $\rightarrow$ 1B parameters
- Progressive resolution training

V-JEPA2-AC: Action-Conditioned World Model

Why V-JEPA 2-AC ?

- Limitation of V-JEPA 2 (action-free):
 - Learns visual representations
 - Cannot predict the effect of **actions**
- Goal: Convert V-JEPA 2 into an **action-conditioned latent**
 - Small amount of interaction data
 - Useful for planning and control
- Output:
 - Predict **future** video representation

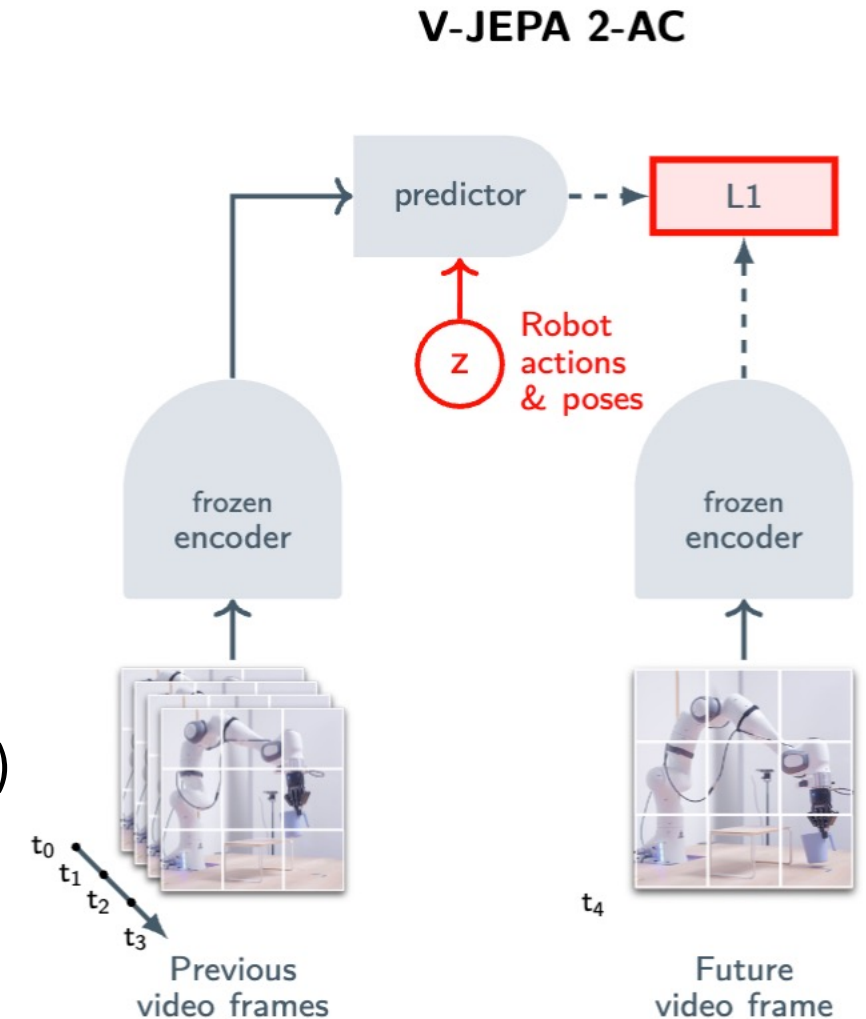
V-JEPA2-AC: Action-Conditioned World Model

Training Data: Droid Dataset

- 62H of unlabeled robot videos
- No reward
- No task labels

Inputs:

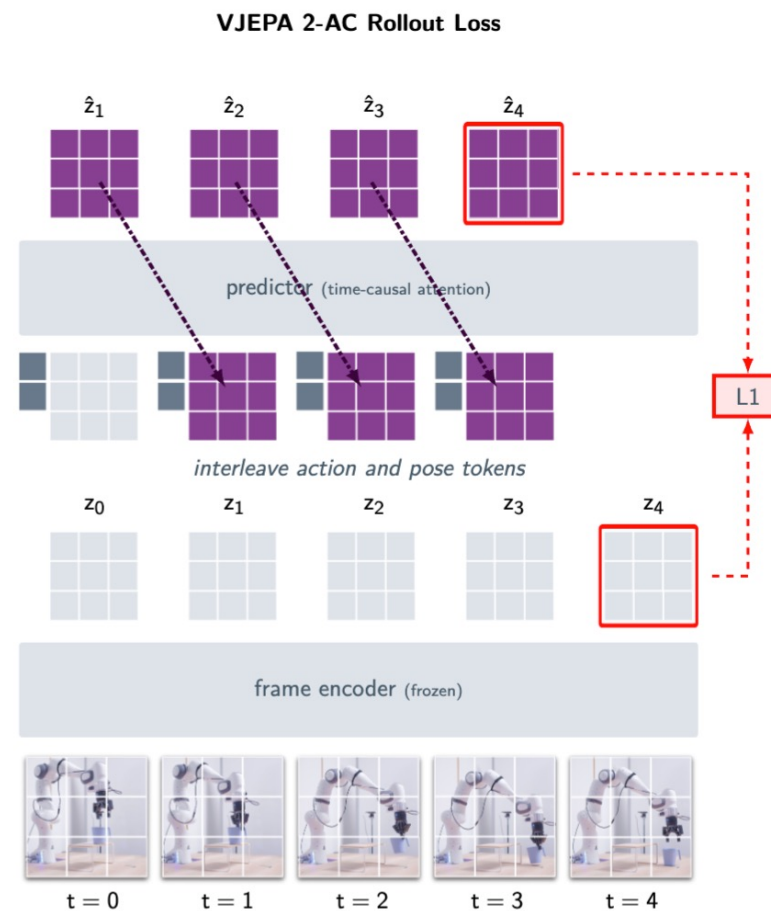
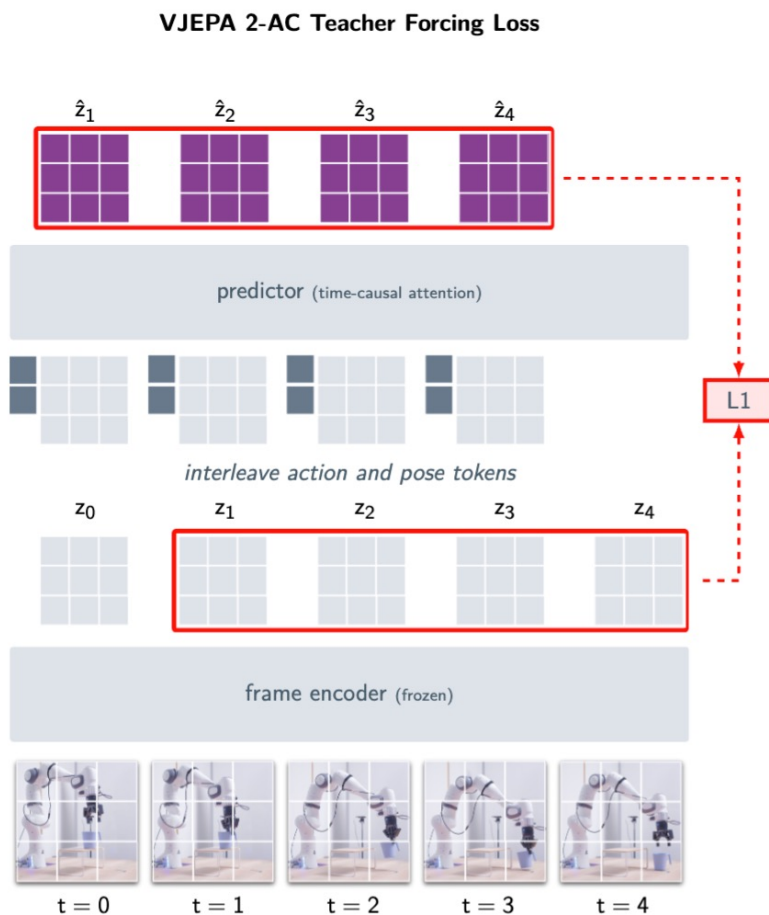
- x_k : Current frame
- s_k : Robot state
 - End-effector state
 - Position (3D) / Orientation (3D) / Gripper state (1D)
- a_k : Action
 - Change in end-effector state



V-JEPA2-AC: Action-Conditioned World Model

Training Objective

- 1) scalar-valued teacher-forcing loss function
- 2) Two-step rollout loss



V-JEPA2-AC: Action-Conditioned World Model

Training Objective

1) scalar-valued teacher-forcing loss function

- Goal: predict next representation ($\hat{z}_{k+1}$)

$$\mathcal{L}_{\text{teacher-forcing}}(\phi) := \frac{1}{T} \sum_{k=1}^T \|\hat{z}_{k+1} - z_{k+1}\|_1 = \frac{1}{T} \sum_{k=1}^T \left\| P_{\phi} \left((a_t, s_t, E(x_t))_{t \leq k} \right) - E(x_{k+1}) \right\|_1$$

- Teacher-Forcing:
 - Input **ground-truth state** as next input (instead of model prediction)
 - Enables stable learning of one-step dynamics

V-JEPA2-AC: Action-Conditioned World Model

Training Objective

2) Two-step rollout loss

- Goal: improve autoregressive rollout during **inference**

$$\mathcal{L}_{\text{rollout}}(\phi) := \|P_{\phi}(a_{1:T}, s_1, z_1) - z_{T+1}\|_1.$$

$P_{\phi}(a_{1:T}, s_1, z_1)$: final predicted state representation

- Motivation:
 - Teacher-forcing uses ground-truth states during training
 - In inference, the model must use its **own predictions**
 - Prediction errors can accumulate

V-JEPA2-AC: Action-Conditioned World Model

Training Objective

- 1) scalar-valued teacher-forcing loss function
- 2) Two-step rollout loss

Overall Training Objective:

$$L(\phi) := \mathcal{L}_{\text{teacher-forcing}}(\phi) + \mathcal{L}_{\text{rollout}}(\phi)$$

- Better long-horizon prediction
- More reliable planning

Action Inference by Planning

From Prediction to Planning

- Training:
 - Current State + Action $\rightarrow$ Predict Future state
- Planning:
 - Current State + Goal State $\rightarrow$ Find Action Sequence

Action Inference by Planning

Energy Minimization

- Goal: Find optimize actions that make predicted future state similar to the goal state
- goal-conditioned energy function:

$$\mathcal{E}(\hat{a}_{1:T}; z_k, s_k, z_g) := \|P(\hat{a}_{1:T}; s_k, z_k) - z_g\|_1$$

z : feature map (k : current, g : goal)

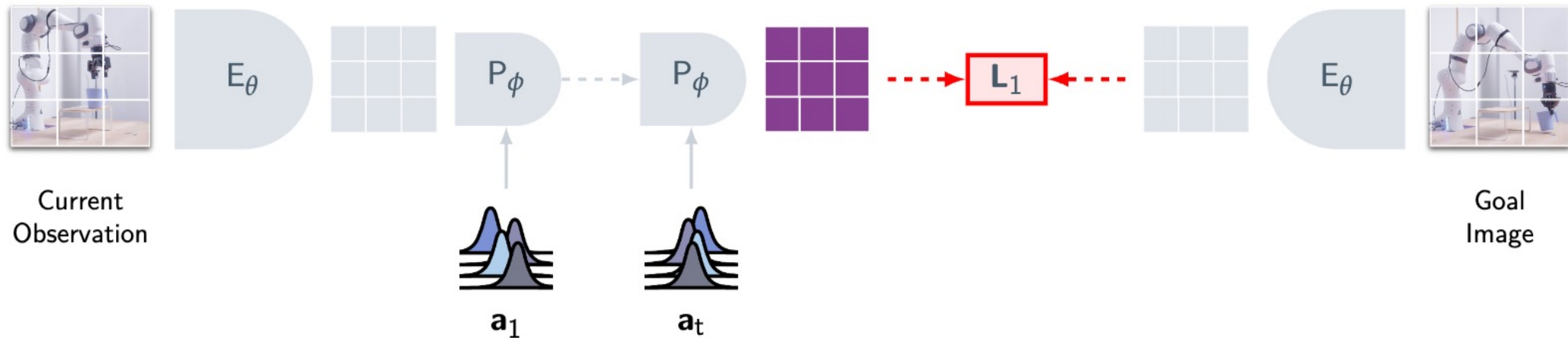
- Action sequence:

$$(a_i^*)_{i \in [T]} := \operatorname{argmin}_{\hat{a}_{1:T}} \mathcal{E}(\hat{a}_{1:T}; z_k, s_k, z_g)$$

Action Inference by Planning

Energy Minimization

- In each Planning Step:
 - 1) Cross-Entropy Method (CEM)
 - Iteratively optimize action sequence through sampling and refinement
 - 2) Receding Horizon Control
 - Execute only the first action and re-plan using new observations



Results: Planning

Planning Performance & Efficiency

Lab 2		Planning Details				Grasp		Pick-&-Place	
Method	#Samples	Iter.	Horizon	Time	Reach	Cup	Box	Cup	Box
Cosmos (Agarwal et al., 2025)	80	10	1	4 min.	80%	0%	20%	0%	0%
V-JEPA 2-AC (ours)	800	10	1	16 sec.	100%	60%	20%	80%	50%

- V-JEPA 2-AC achieves higher success rates across robot manipulation tasks
- Predicting in latent representation:
 - More effective
 - More computationally efficient

Conclusion

- Self-Supervised video pretraining learns strong visual representations
- V-JEPA 2 supports **understanding and prediction** across diverse tasks
- V-JEPA 2-AC enables **planning** and zero-shot robot control
- Demonstrates a unified framework for understanding, prediction and planning from video

Future Works

- Long-horizon Planning
- Language-based Goal Specification
- Scaling